SECTION WA

Wheels, Axles & Chains

Table of Contents

IDLER SHAFT INSPECTION - MAINTENANCE SCHEDULE (FRONTIER MODELS)..... WA-18
REMOVING THE IDLER SHAFT (FRONTIER MODELS) WA-19
INSTALLING THE IDLER SHAFT (FRONTIER MODELS) WA-20
REMOVING THE IDLER SHAFT (AURORA MODELS)..... WA-23
INSTALLING THE IDLER SHAFTS (AURORA MODELS) WA-24
REMOVING AN AXLE/SPROCKET/BEARING (WITHOUT BEARING EXTENSION) WA-25
INSTALLING AN AXLE/BEARING/SPROCKET (WITHOUT BEARING EXTENSION)..... WA-26
REMOVING AN AXLE/BEARING/SPROCKET (WITH BEARING EXTENSION) WA-28
INSTALLING AN AXLE/BEARING/SPROCKET (WITH BEARING EXTENSION) WA-29
DRIVE CHAIN AND SPROCKET ORIENTATION (8X8 MODELS)..... WA-31
DRIVE CHAIN AND SPROCKET ORIENTATION (6X6 MODELS)..... WA-32
REMOVING THE INNER BEARING..... WA-33
INSTALLING THE INNER BEARING..... WA-33

General Instructions

WARNING

Detailed information on standard workshop and safety procedures, and general servicing operations is not included in this manual, which has been prepared to assist qualified service personnel. ARGO assumes no responsibility or liability for PERSONAL INJURY or VEHICLE DAMAGE which results from any servicing procedure performed, including those instructions outlined in this manual.

WARNING

Improperly adjusted idler chains or poorly aligned idler shaft assemblies can result in personal injury to the operator, extensive damage to brake disc sprockets and idler shaft assemblies, and may lead to a costly and inconvenient break down off road. Please ensure that the axle shafts are correctly aligned and that the drive chains and chain tensioner system are in good working condition.

WARNING

Improperly aligned axle shafts or incorrect chain tension can result in extensive damage to drive chains and axle sprockets, and may cause wear on the vehicle frame and floorpans. Please ensure that axle shafts are correctly aligned and that the drive chains and chain tensioner system are in good working order.

Lubrication Information

General

The following parts and components require regularly scheduled lubrication to prevent premature wear and replacement.

1. Idler Chain (Frontier Models Only)
2. Drive Chains
3. Bearings

Use the recommended lubricants listed in this section and carefully observe the recommended lubrication intervals in the General Information section of this service guide.

Drive Chain Lubrication

The ARGO vehicle is equipped with roller chains to each axle. Lubricate the chains every 10 hours or more frequently in dirty or wet conditions with an Aerosol Chain Lube (or use ARGO brand Chain Lube (part no. 130-100).

WARNING

Never use gasoline as a cleaning solvent on drive chains. Gasoline is extremely flammable and can explode if ignited, causing serious personal injury.

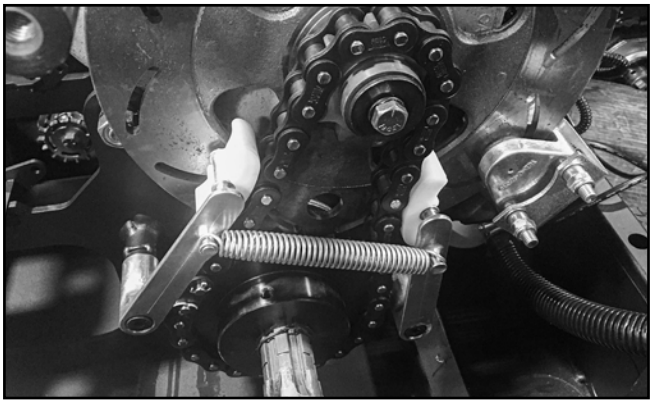
Idler Chain (Frontier Models Only)

Frontier Models are equipped with 2 idler chains to transfer power from the output shafts of the transmission to the drive chains through a series of sprockets. They are located on each side of the transmission.

A loose or improperly adjusted idler chain can result in damage to the chain or sprockets. During the new vehicle break-in period, check the idler chain adjustment before operating the vehicle and after the first 2 hours of operation. After the initial break-in period, check the idler chain adjustment every 50 hours. An indication of loose idler chains is a loud banging noise when the vehicle is turned.

Idler Chain Adjustment - Self Adjust Idler Chain (Frontier)

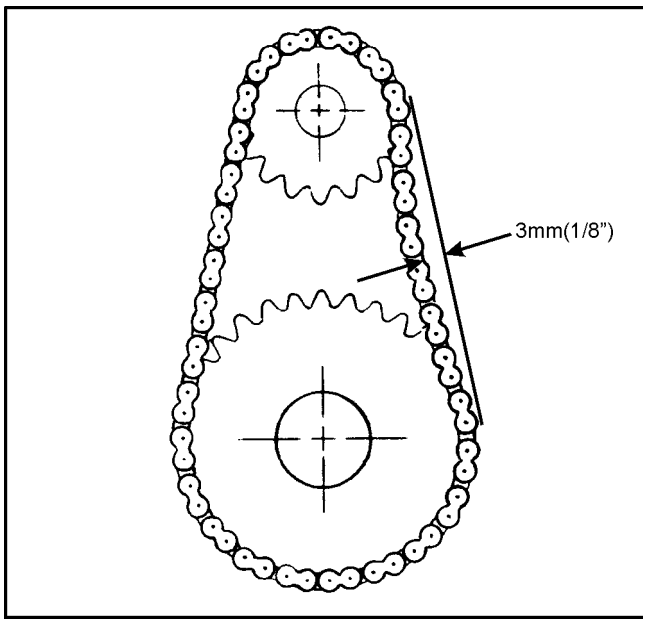
Idler chains are self-adjusting, spring loaded. As the chain stretches, tensioner blocks squeeze the chain to remove the slack through tensioner arms.



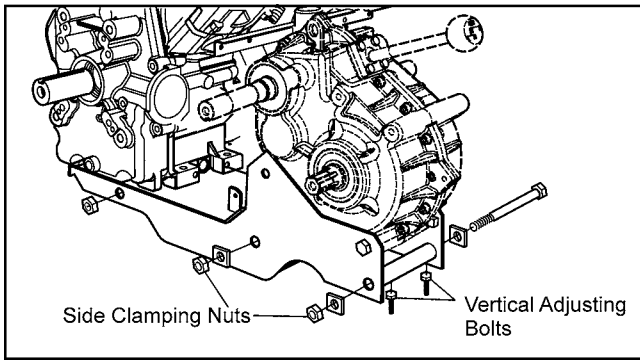
Idler Chain Adjustment - Manual Adjust (Frontier)

A loose or improperly adjusted idler chain can result in damage to the chain or sprockets. During the new vehicle break-in period, check the idler chain adjustment before operating the vehicle and after the first 2 hours of operation. After the initial break-in period, check the idler chain adjustment every 50 hours. An indication of loose idler chains is a loud banging noise when the vehicle is turned.

To check the idler chain adjustment, push the slack side of the chain and measure the amount of chain deflection. Adjust the idler chain tension if deflection is more than 3mm (1/8").

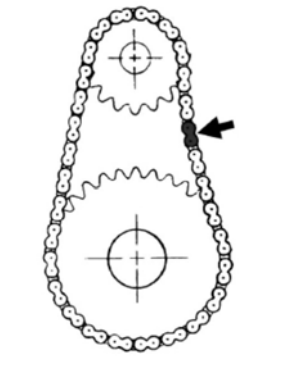


1. Remove the firewall from the vehicle.
2. Loosen the 2 left side clamping nuts with a 15/16" socket wrench.
3. Turn the vertical adjustment bolts counter-clockwise to raise the power pack and tighten the idler chains. The idler chains are properly adjusted when the deflection measures 3 mm (1/8").
4. Tighten the 2 left side clamping nuts securely. **Torque to 80ft./lbs.**

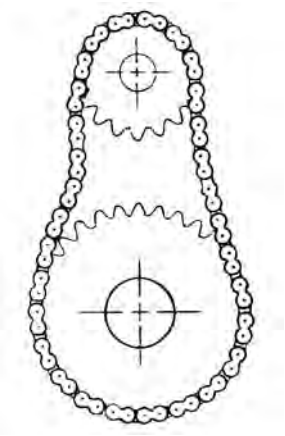


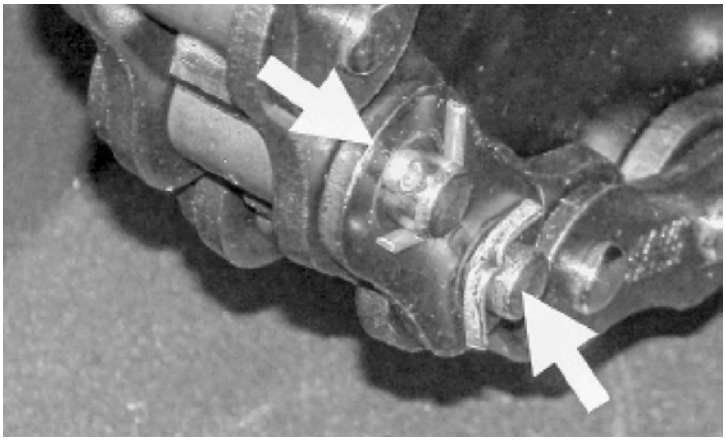
Removing The Idler Chains Manual Adjust (Frontier)

1. The idler chain is tensioned by two chain tensioner bolts located at the back of the power pack frame.
2. Remove the vehicle firewall.
3. Loosen the 2 left side clamping nuts with a 15/16" socket wrench.
4. Turn the vertical adjustment bolts clockwise to lower the power pack and loosen the idler chains.
5. Place the gearshift in neutral and roll the vehicle's tires until the connecting link of the idler chain is easily accessible.



6. Remove the cotter pins from the connecting link. Remove the outside plate and tap out the connecting link.
7. Remove the idler chain from the vehicle.





Installing The Idler Chain - Manual Adjust(Frontier)

1. Install the chain over the brake disc sprocket and the idler shaft sprocket.
2. Pull the ends of the chains together and insert the connecting link.

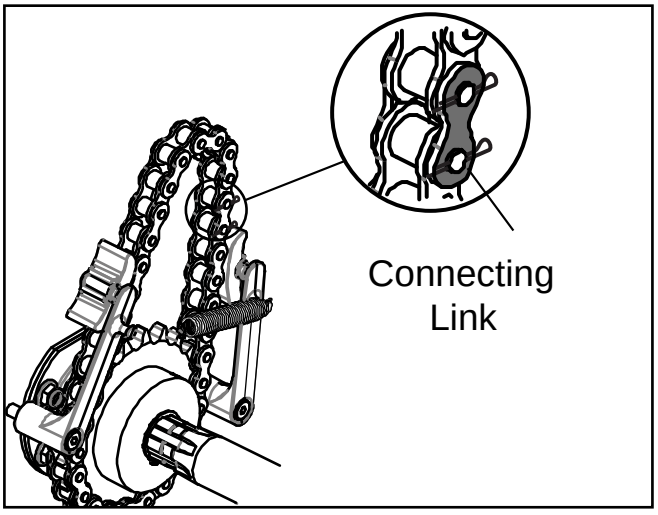
⚠ NOTE

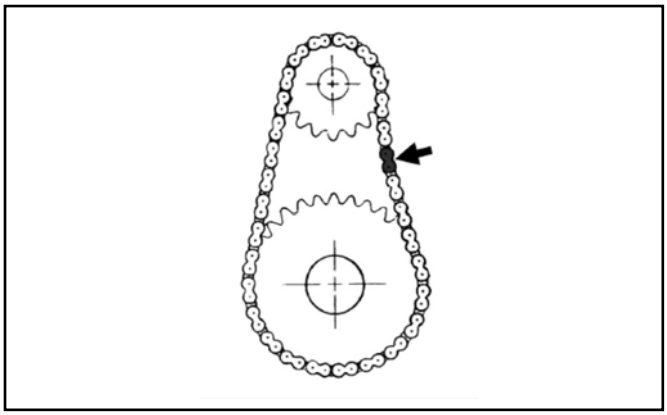
Make sure the idler chain adjustment bolts are at their lowest level. Use a pair of modified 7R Vice Grips to hold the ends of the chain together while inserting the connecting link. There may be no slack in the idler chain, making installation of the connecting link difficult without this tool. Modified Vice Grips can be ordered from your ARGO dealer (ARGO Part No. 658-08).

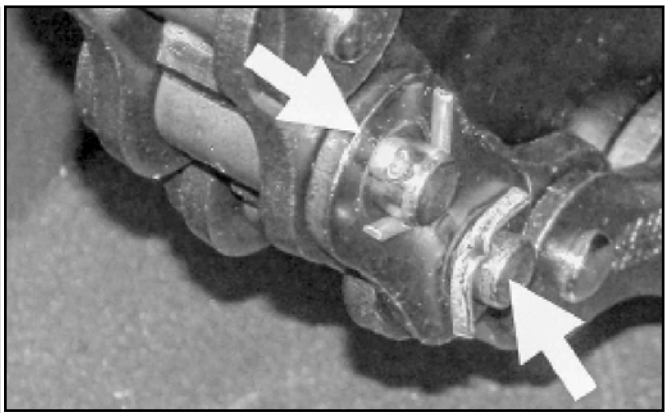
3. Install the outside plate and secure with the cotter pins. Always use new cotter pins.
4. Perform the servicing procedure, **Idler Chain Adjustment.**
5. Reinstall the firewall.

Removing The Idler Chain-Self Adjust (Frontier)

1. The idler chain is fitted with a chain tensioner system. Remove the spring between the tensioner arms and the lay the blocks back away from the chain.
2. With the vehicle raised off the floor, place the gearshift in neutral and roll the vehicle until the connecting link is positioned in an accessible spot.





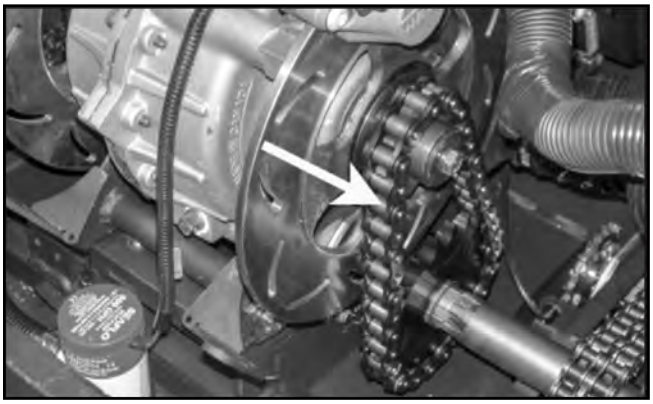


3. Remove the cotter pins from the connecting link. Remove the outside plate and tap out the connecting link.
4. Remove the idler chain from the vehicle.

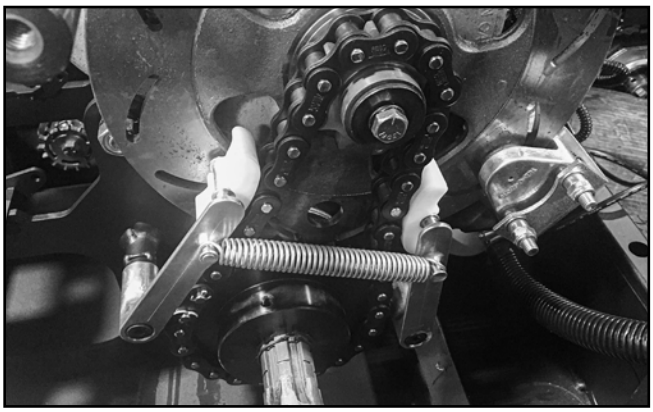
Installing The Idler Chain-Self Adjust (Frontier)

1. Install the chain over the brake disc sprocket and the idler shaft sprocket.
2. Pull the ends of the chains together and insert the connecting link.
3. Install the outside plate and secure with the cotter pins. Always use new cotter pins.
4. Reinstall the tensioner spring between the idler chain tensioner arms.
5. Reinstall the firewall.

Idler Chain Lubrication (Frontier Models Only)

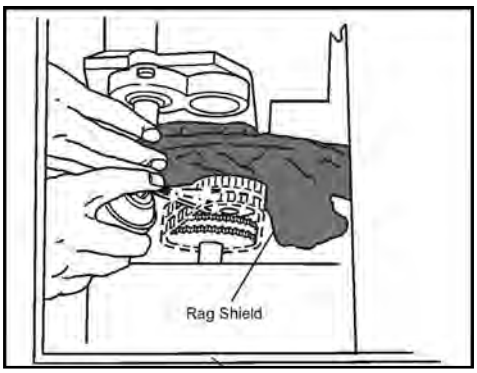


Frontier Idler Chain W/O chain tensioner



Frontier Idler Chain W/chain tensioner

When applying the chain lube, protect the brake discs with a rag or simple cardboard shield. DO NOT SPRAY CHAIN LUBE ON THE BRAKE DISCS OR PADS. Roll the vehicle so that all of the idler chain is accessible for lubrication.



CAUTION

Do not use regular oil or grease to lubricate the idler chains. Regular oil or grease will be thrown off the idler chains during normal operation, and contaminate the brake pads or discs. When these components become contaminated, brake effectiveness can be compromised or complete brake failure can occur.

Regular maintenance of the drive chains is essential to both good vehicle performance and personal safety. Poorly maintained chains can result in rusted, corroded or seized up chains, damaged sprockets, damaged frame, and costly repair bills.

1. Every 25 hours of vehicle operation perform the servicing procedure, **Drive Chain Inspection**.
2. Every 10 hours of vehicle operation perform the servicing procedure, **Drive Chain Take-Up System Inspection**.
3. Every 10 hours of vehicle operation perform the servicing procedure, **Drive Chain Lubrication.**
4. Every 100 hours of vehicle operation perform the following servicing procedures:

- Drive Chain Inspection
- Chain tensioner Inspection

Outer Axle Flange & Bearing Lubrication

CAUTION

Do Not Use High Pressure Pneumatic Greasing Equipment

Outer Axle Bearing Lubrication

Each outer axle flange is equipped with one grease nipple. This grease nipple supplies grease directly to the outer axle bearing . It also requires re-greasing every 25 hours of operation or before the vehicle is taken out of service for any extended period. Use a pistol grip type grease gun to avoid dislocating the bearing seals due to excessive grease pressure.
See NOTE following page.

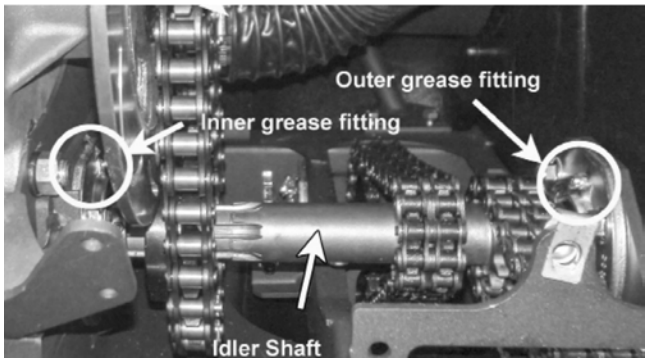


NOTE

To promote regular maintenance of important Argo components, ARGO has provided an access hole through each rim and hub for ease of bearing lubrication.

Idler Shaft Bearing Lubrication (Frontier Models Only)

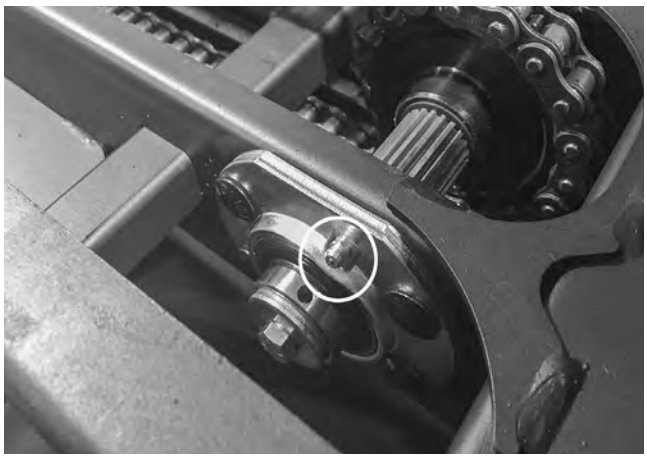
Left and right hand side inner and outer idler shaft bearings are fitted with a right angled grease fitting. With the front floor pan removed, the left side inner idler shaft grease fitting is accessible at the bottom of the bearing flange. The right side inner idler shaft grease fitting is located at the top of the flange.



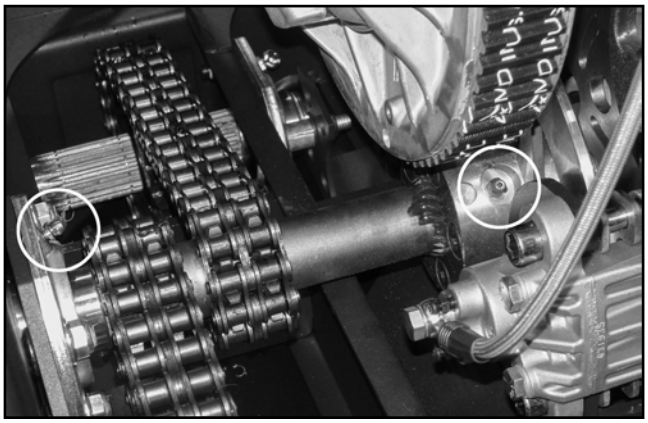
Both outer idler shaft bearing grease fittings are located at the top of the flange. All of these can be accessed conveniently with a grease gun fitted with a flexible extension head. Grease with a small amount of a lithium based, NLGI #2 or 3 mineral oil based grease, (such as Shell Alvania #3) or ARGO Brand Multi-Purpose Grease Part No. 130-105). Apply every 50 hours of operation, if vehicle has been used in water for extended periods of time or whenever major maintenance is performed on the vehicle.

Inner Axle Bearing Lubrication

The inner axle flanges are equipped with a grease nipple Lubricate the bearings with a small amount of a lithium based, NLGI #2 or 3 mineral oil based grease, (such as Shell Alvania #3) or ARGO Brand Multi-Purpose Grease Part No. 130-105). Apply every 50 hours or before the vehicle is taken out of service for any extended period. Dirt, dust and exposure to water will accelerate this servicing to less than 50 hour intervals. Only a small amount of grease is required.



Output Shaft Coupler Lubrication (Aurora Models Only)



Output shaft spline couplers are equipped with a grease fitting to allow lubrication to the spline of the shaft and coupling connector. Lubricate every 25 hours with a lithium based, NLGI #2 or 3 mineral oil based grease, (such as Shell Alvania #3 or ARGO Brand Multi-Purpose Grease Part No. 130-105). Wipe off excess.

Drive Chains

Roller chain "stretch" results from wear to the chain pins and bushings because of the loss of lubricant.

To prevent sprocket damage and unnecessary breakdowns, replace the chains when:

- the chain tensioner can no longer take up the chain slack.
- the chain is rubbing on a frame cross member.
- the chain is seized due to rust and lack of lubrication.
- the chain climbs the sprocket teeth, especially noticeable when turning.

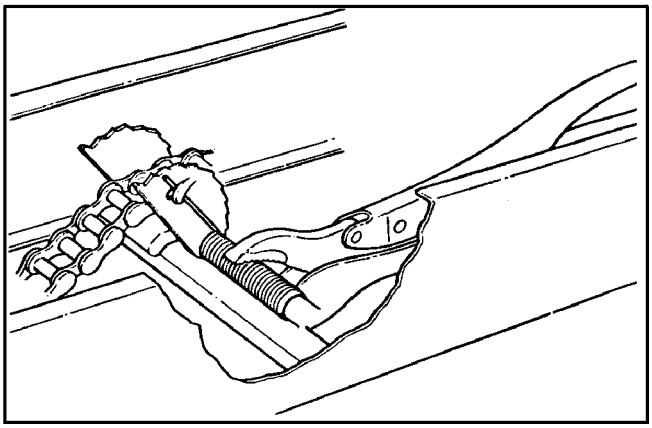
Removing a Drive Chain

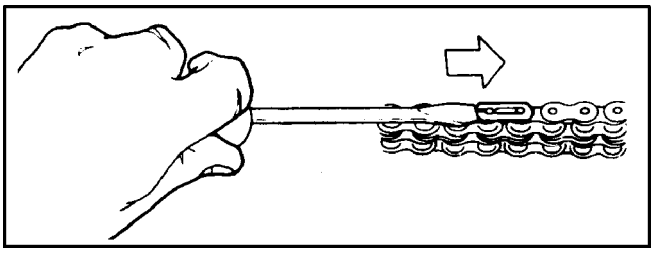
1. Place the gearshift in the N (neutral) position.
2. Remove the floor pans.

Semi-Self Adjust Chain Tensioners:

a. Turn the tensioner cam assembly in the direction which winds up the torsion spring and push the assembly down as close as possible to the cam follower block in the bottom of the frame. Secure it in this position with a Vice-Grip 10CR.

3. Roll the vehicle until the connecting link on one of the chains is visible.
4. Remove the spring clip from the connecting link. Remove the outside plate and tap out the connecting link. The inside plates will be released when the connecting link is removed





Scissor Style Chain Tensioners:

b. Release the spring torsion on the scissor style chain tensioner system.





5. Proceed to previous steps 3 and Step 4.
6. Remove the chain from the vehicle.

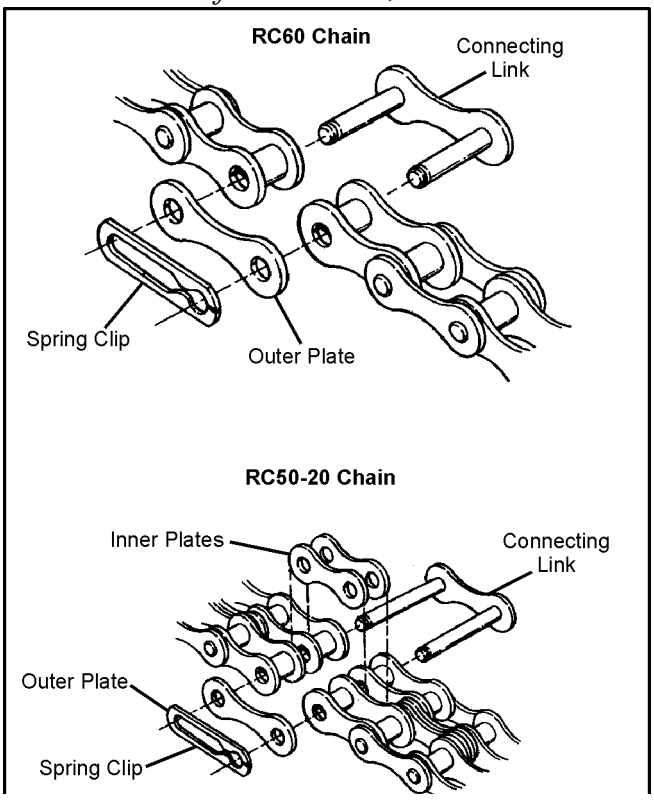
Installing a Drive Chain

1. Position the drive chain over the slider block and around the drive sprockets.
2. Pull the ends of the chain together and insert the connecting link. When connecting the RC50-2 chain, insert the inside plates before tapping the connecting link into position.

⚠ NOTE

Use a pair of modified 7R Vice Grips to hold the ends of the chain together while inserting the connecting link. Some drive chains have no slack, and replacement of the connecting link is difficult without this tool. Modified Vice Grips can be ordered from ARGO, Part No. 658-08.

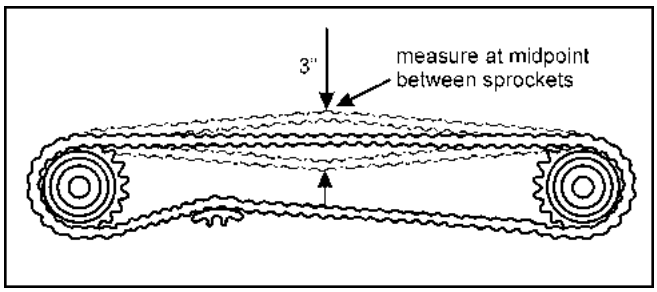
3. Replace the outside plate and spring clip. The open end of the clip should face rearward when it is on top of the chain.
4. Remove the vice-grips securing the cam assembly in its lowest position.



Drive Chain Take-Up System Inspection (Semi-Self Adjust)

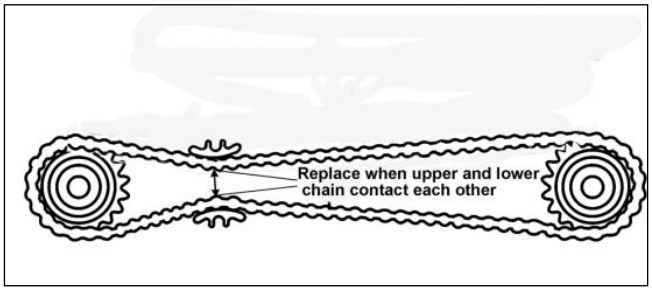
The semi-self adjust chain tensioning system consists of a torsion spring loaded cam assembly with a slider block which takes up the slack on the bottom side of all but the front final drive chains. As the chain wears, the chain tensioning mechanism adjusts semi-automatically. Under most conditions, the tensioner cam assembly will move to the next step of adjustment simply due to normal drive system dynamics. Sometimes, however, the cam assembly can bind due to debris caught in the area.

IMPORTANT



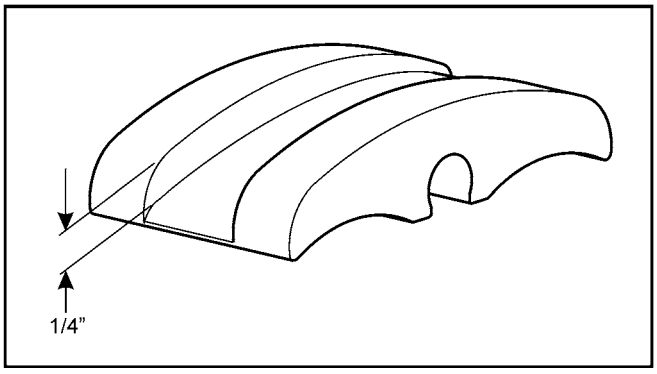
Check that the cam assembly is progressing properly. Check for proper chain tensioner operation every 10 hours of vehicle operation, when the drive chains are being lubricated. Each step of the cam takes up about 3 inches of chain slack

Drive Chain Take-Up System Inspection Scissor Style



The scissor style chain tensioning system consists of a torsion springed upper and lower slider block. As the chain wears, the chain tensioning mechanism adjusts automatically by squeezing the chain from both top and bottom. Replace chains when upper and lower chain start to contact each other.

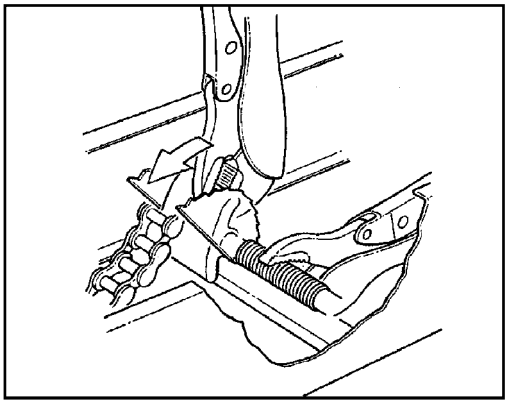
Chain Tenioner Slider Block Inspection

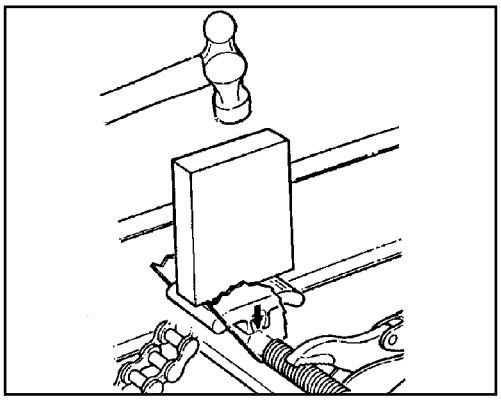


Each chain tensioner system consists of either a single UHMW slider block (situated below the lower chain) or two slider blocks (situated above and below the drive chain). Inspect the UHMW slider blocks for wear every 50 hours. Replace the blocks when the wear groove measures 1/4" (6mm).

Replacing A Slider Block (Semi-Self Adjust)

1. Remove the floor pans.
2. Turn the tensioner cam assembly in the direction which winds up the torsion spring and push the assembly down as close as possible to the cam follower block across the bottom of the frame channels. *See following page.* Clamp it in this position with a Vice-Grip 10CR or similar plier and perform the servicing procedure, **Removing The Drive Chain.**
3. With pliers, pry the slider block off the cam assembly.
4. Place a new slider block over the shaft of the cam assembly.
5. Using a piece of wood (or similar material) pressed against the top of the slider block, carefully hammer the piece of wood so the slider block snaps onto the cam assembly shaft.





6. Perform the servicing procedure, **Installing The Drive Chain**.
7. Pull up on the cam assembly to allow it to take up as much chain slack as possible.
8. Replace the floor pans.

Replacing A Slider Block (Scissor Style Chain Tensioner)

1. Remove the floor pans.
2. Release the spring torsion on the scissor style chain tensioner system.



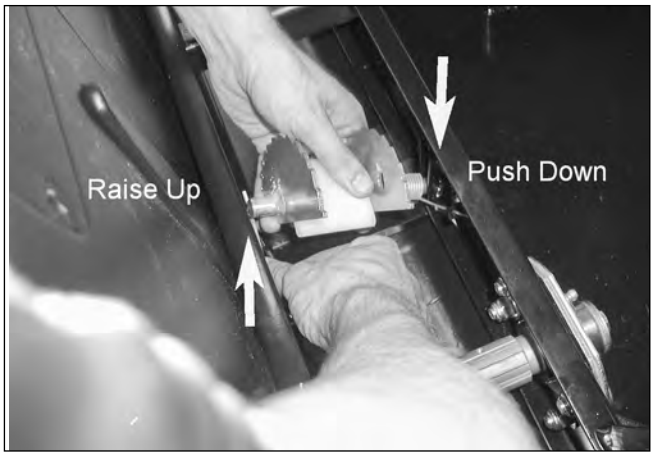


3. With pliers, pry the slider block off each of the change tensioner arms.
4. Install the new slider blocks by snapping each one on to the chain tensioner arms using the pliers.
5. Reconnect the tension spring between each tensioner arm.

Disassembling The Chain Tensioner System (Semi-Self Adjust)

1. Perform the servicing procedure, **Removing The Drive Chain.**
2. With the chain removed, unfasten the clamp holding the chain tensioner system and let it unwind to its top most position. Unfasten the spring at the cam's tab. To pull the tensioner assembly free of the frame guides, push the long shaft side of the cam assembly down to the bottom of the guide. At the same time raise the opposite side up and out of the guide.





3. Slip the one torsion spring off the cam, and remove the second torsion spring from the frame assembly by removing the cotter pin that secures it.



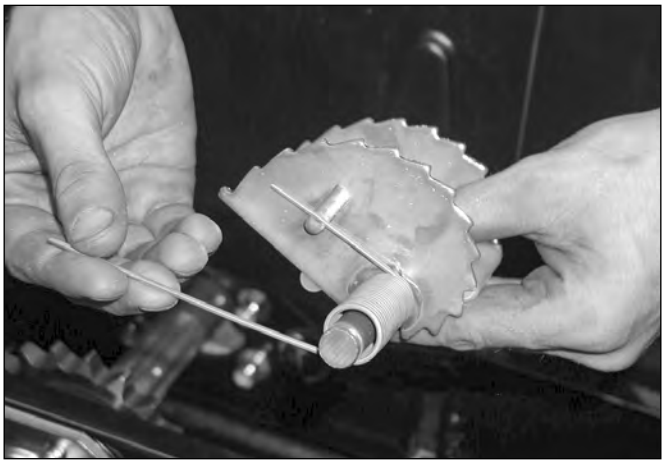


Assembling the Chain Tensioner System (Semi-Self Adjust)

1. Place the tensioner block onto a work-bench and position the slider block over the shaft of the assembly.
2. With a soft faced mallet, tap the slider block to snap it on to the shaft.
3. Slip a torsion spring over the welded bushing in the frame, and secure with a cotter pin.
4. Slip the second torsion spring on to the shaft of the tensioner cam assembly.

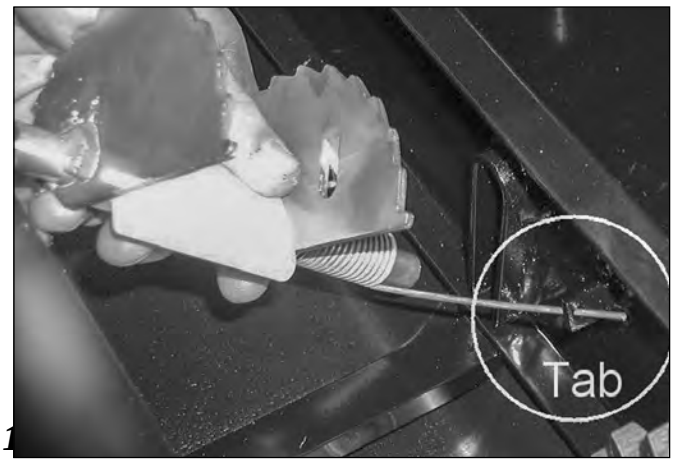


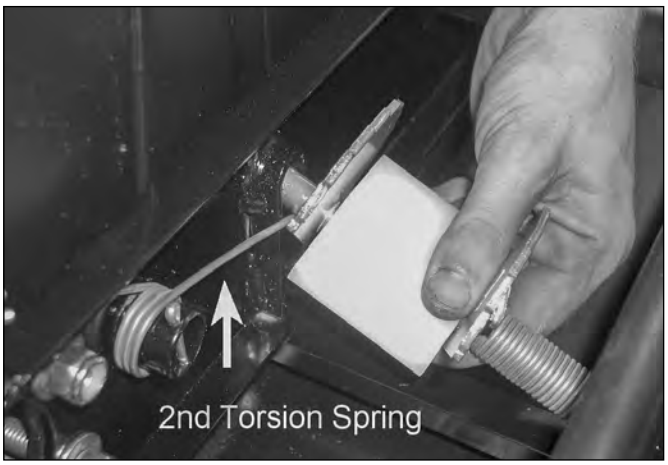




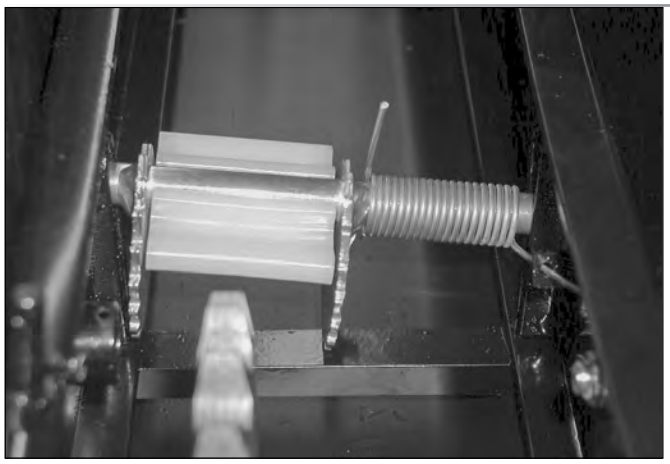
5. Insert the extended length of the torsion spring, assembled to the cam, into the hole of the welded tab on the frame. At the same time, place the same side of the cam's shaft into the tensioner systems guide. *See following page.*

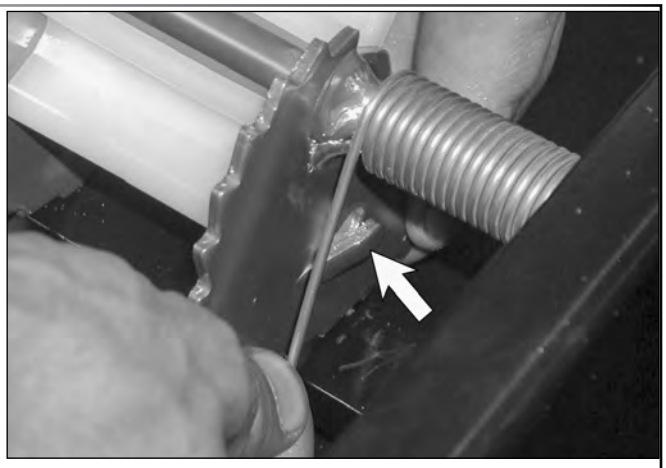
6. Push the previously inserted spring side of the cam to the bottom of the guide and raise the opposite side to place over top the extended length of the second torsion spring on the opposite side.

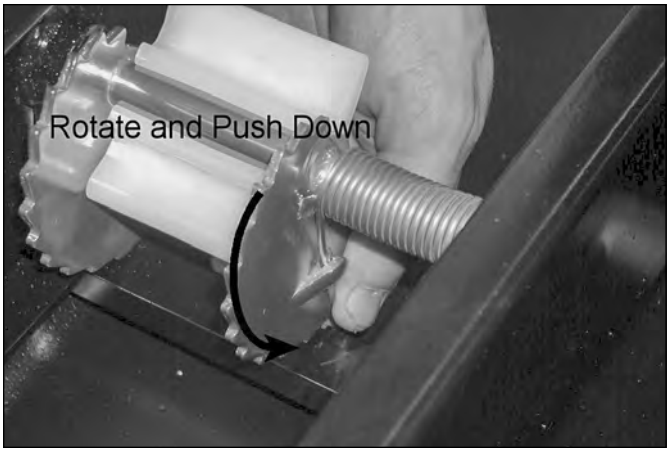


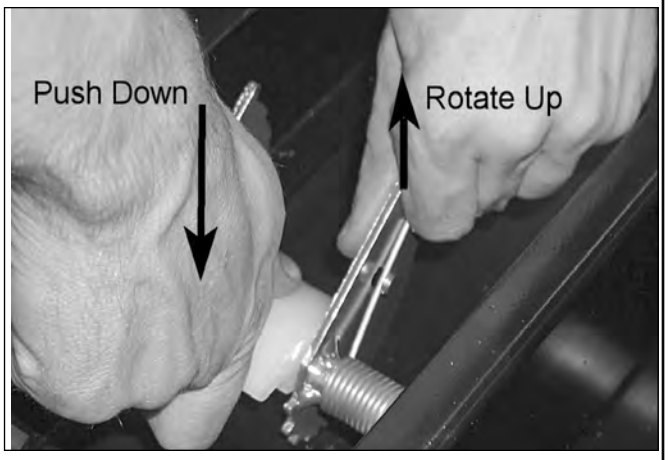


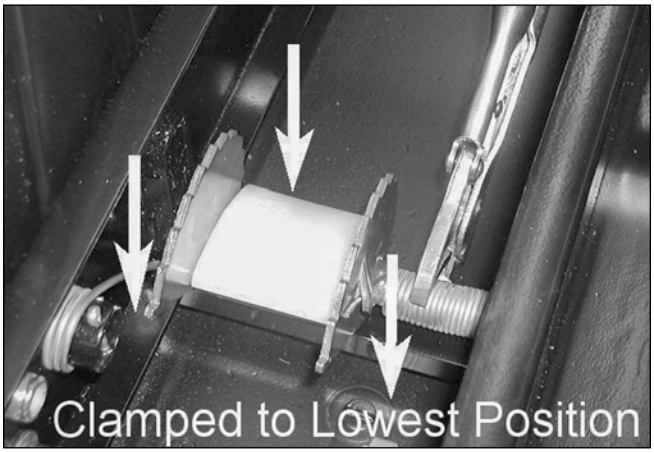
7. Allow the assembly to sit in the up most position.
8. Assemble the short stem of the torsion spring on the cam's shaft, into the tab on the cam.
9. Turn the cam assembly in the direction which winds up the torsion spring at the same time pushing the assembly down as close as possible to the cam follower block across the bottom of the frame channels. Clamp it in this position until the chain is reinstalled. *See following page.*











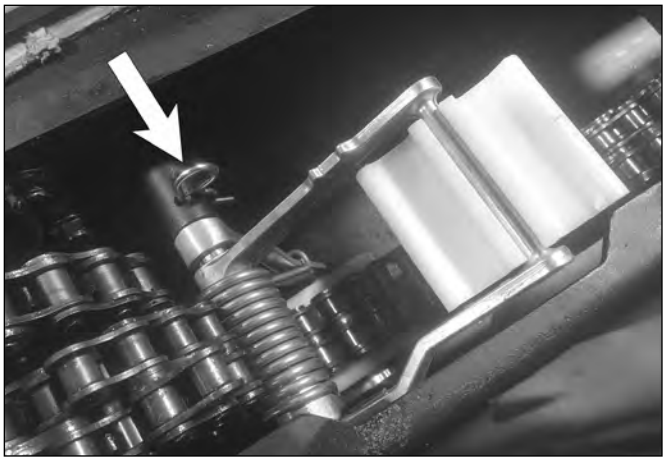
10. Perform the servicing procedure, **Installing the drive chain.**

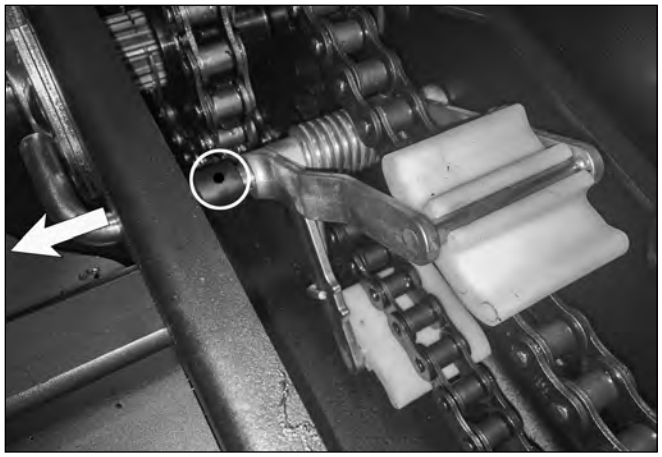
Disassembling The Chain Tensioner System (Scissor Style Chain Tensioner)

1. Release the spring tension on the tensioner arms and fold the upper tensioner block back out of the way. Perform the servicing procedure, **Removing A Drive Chain.**
2. Locate and remove the Hair Pin from the Tensioner Pin.

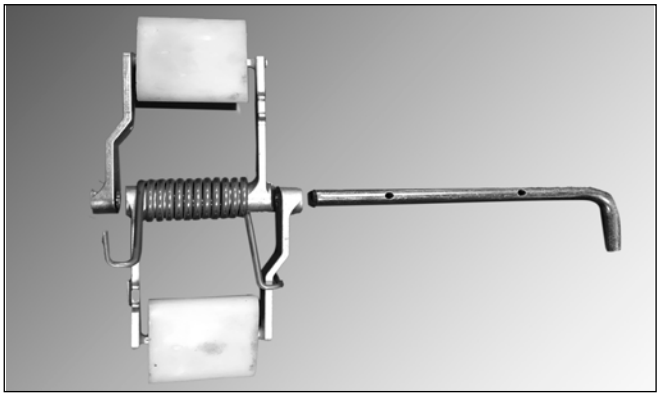
NOTE

Depending on the location and orientation of the chain tensioner the hair pin will be located inboard at the inner frame rail or outboard at the outer frame rail.

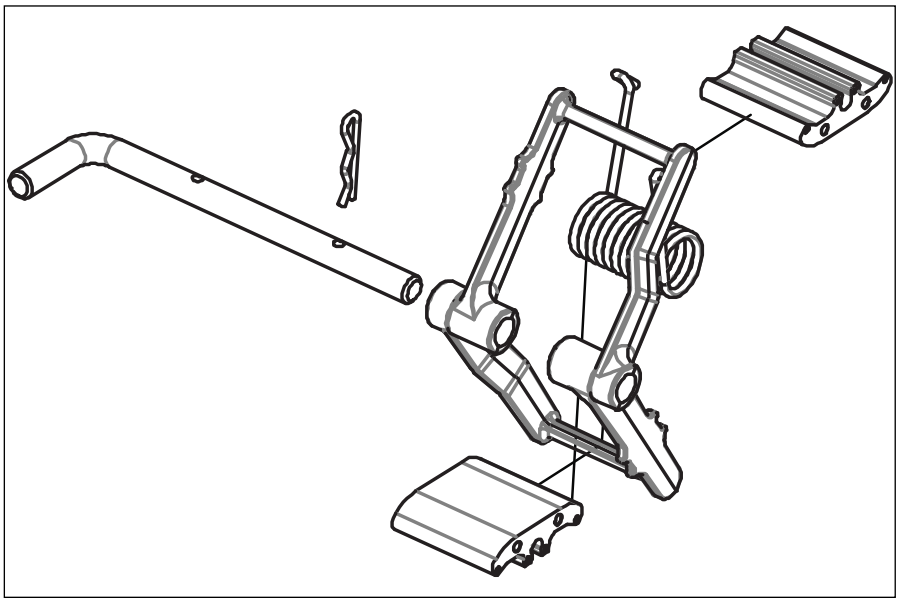




3. Pull the pin from the tensioner arms and remove the arms and spring assembly from the vehicle
4. Lay out the components on a clean work surface.
5. Perform the servicing procedure, **Chain Tensioner Slider Block Inspection**
6. Perform the servicing procedure, **Replacing A Slider Block.**

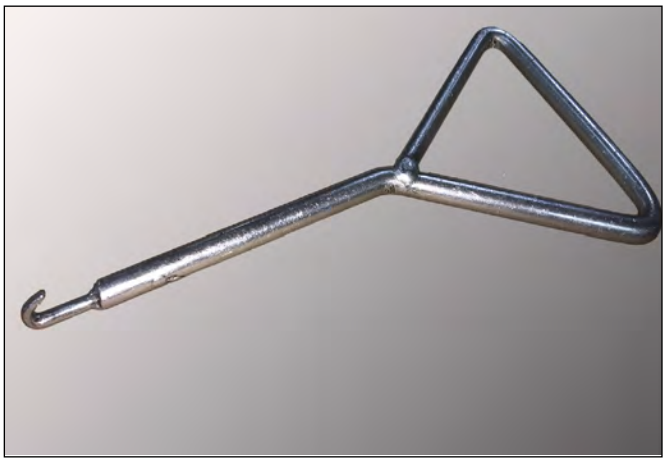


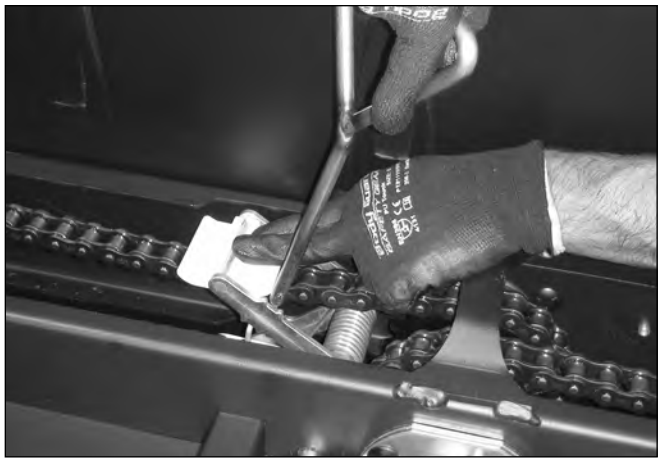
Assembling The Chain Tensioner System (Scissor Style)



1. Assemble the upper and lower chain tensioner arms together along with the centre tensioner spring as in the photo above.
2. Install the assembly between the frame channels ensuring the lower chain is on top of the lower slider block.

3. Insert the Tensioner Pin through the arm assemblies and into the bushing weldment on the frame.
4. Secure with the Hair Pin.
5. Pull up the tensioner spring and clip it over the tensioner arm. Employ a hook style tool to grab the spring end and ease installation.
6. Perform the servicing procedure, **Installing The Drive Chain.**





Idler Shaft Inspection (Frontier Models)
Maintenance Schedule - Inspect every 30 hours of vehicle operation.

CAUTION

Damage to the idler shaft and sprockets can result from loose idler chains, couplers or sprocket misalignment. Please inspect regularly.

1. Perform the servicing procedure, **Removing the Firewall**
2. Inspect the sprockets on the idler shaft thoroughly for:

- cracks in the casting
- worn, broken or missing teeth

If any of these conditions are noted replace drive sprocket.

IMPORTANT

If any wear is noted on the Frontier models idler shaft or any of it's components, it is advisable to check the brake disc drive sprocket to ensure that the misalignment has not caused any damage to those teeth as well

Inspect the idler shaft thoroughly for:

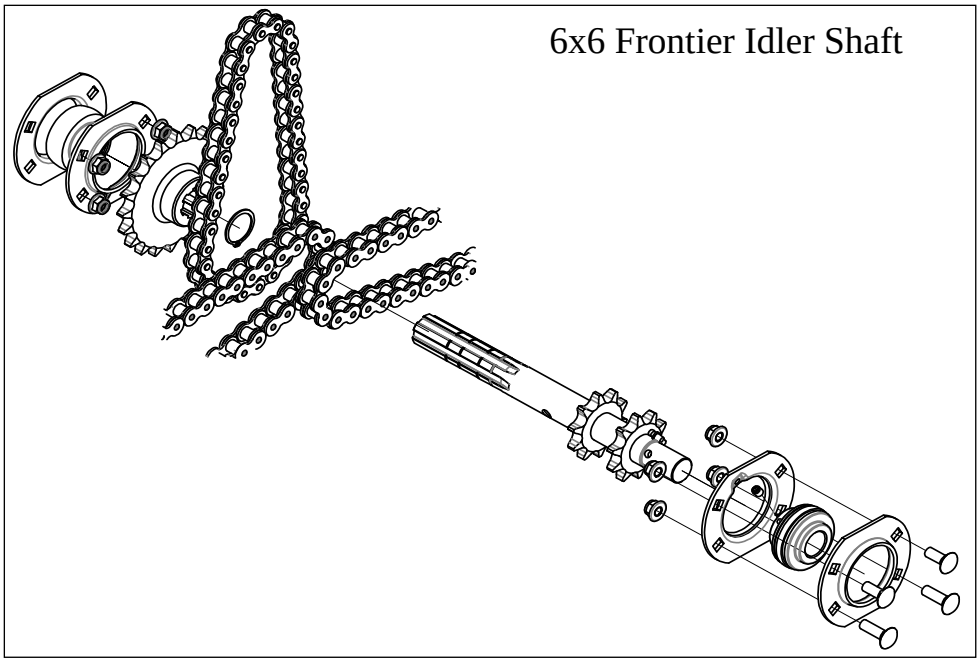
- bends
- rust or corrosion

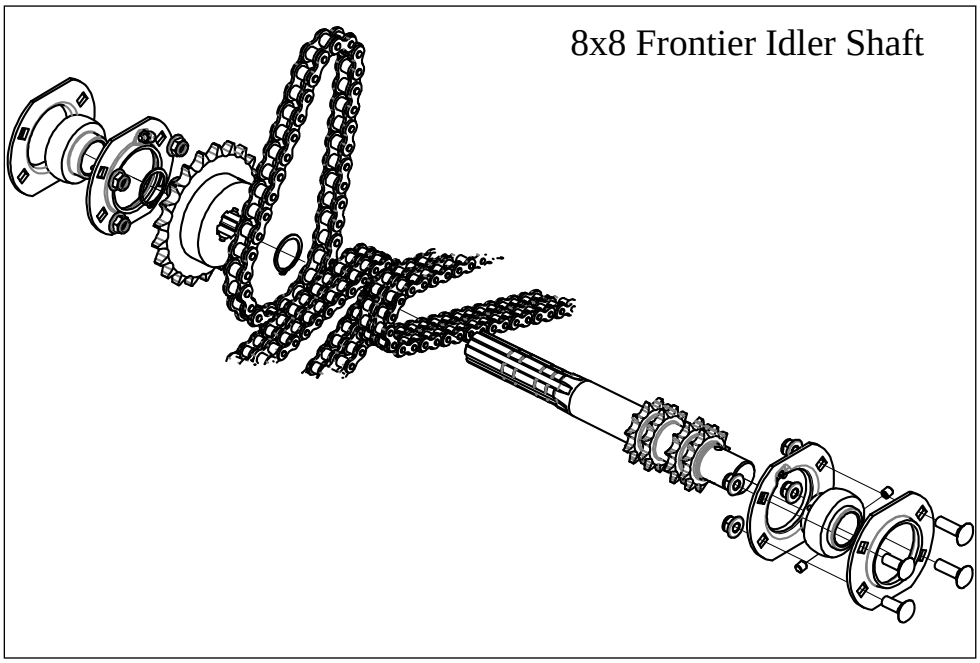
If any of these conditions are noted, replace the idler shaft immediately.
Inspect all the idler shaft fittings for:

- seized or worn bearings
- loose set screws at the bearings
- loose inner or outer flanges
- broken or cracked retaining rings

Replace any seized or worn bearings. Perform the servicing procedure, **Removing the Idler Shaft** and **Installing the Idler Shaft.**

Removing the Idler Shaft (Frontier Models)





1. Perform the servicing procedure, **Removing the Firewall.**
2. Perform the servicing procedure, **Removing the Front Floor Pan.**
3. Perform the servicing procedure, **Removing the Driven Clutch**, (Left side idler shaft only). *This optional step allows easier access to the idler shaft*).
4. Perform the servicing procedure, **Removing the Hydraulic Brake Caliper - Frontier Steering Brakes.**

IMPORTANT

You do not have to disconnect the brake line from the hydraulic caliper.

5. Perform the servicing procedure, **Idler Chain Removal.**
6. Perform the servicing procedure, **Removing the Brake Disc.** ***(****Optional, if you find it easier to access the idler shaft inner flange hardware).*
7. Perform the servicing procedure, **Drive Chain Removal.** Remove the front drive chain connecting the front wheel to idler shaft sprocket, and midfront chain connecting mid front wheel to idler shaft sprocket.
8. Locate the 2 set screws on the inner and outer bearings and remove.
9. With the set screws removed, pry the idler axle until it slides freely by hand within the inner and outer bearing.
10. Release and pull the outside retaining ring back from the sprocket as far as possible and slide the sprocket up against the retaining ring.
11. Remove the hardware securing the flanges of the inner and outer bearings.
12. Push the Idler Shaft assembly into the inner bearing until it is free of the outer bearing.
13. Tilt upwards and remove from vehicle frame.

Installing the Idler Shaft (Frontier Models)

Assembling the Idler Shafts (Frontier Models)

Apply some LOCTITE anti-sieze compound to each end of the shafts.

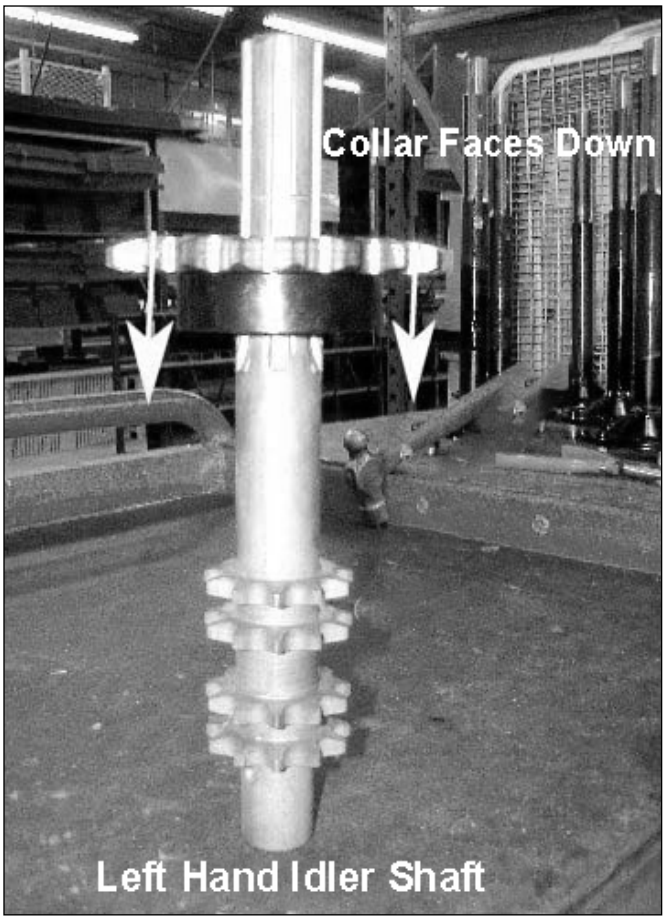
Left & Right Hand Side Idler Shafts

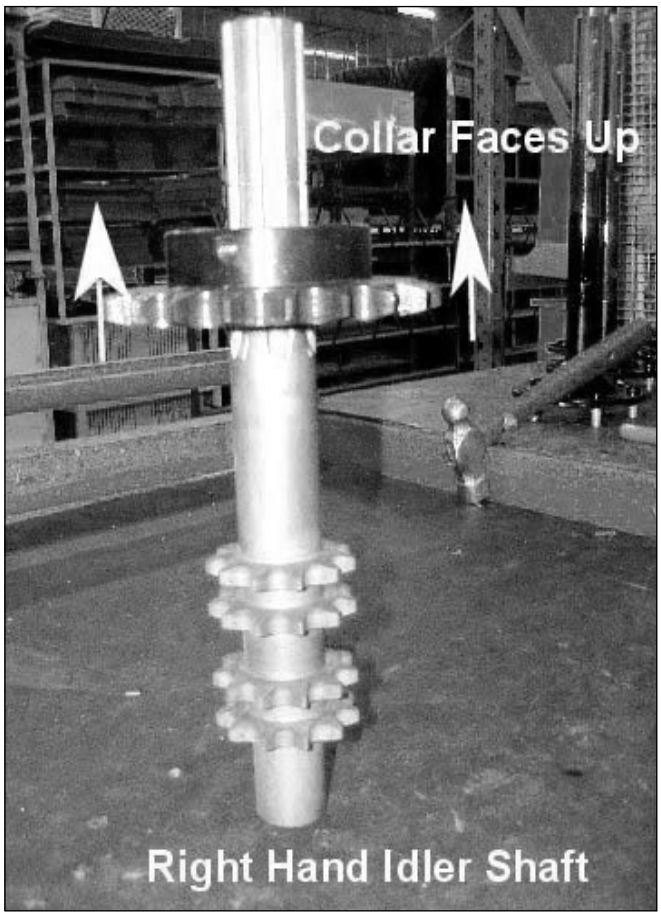
NOTE

8x8 idler shafts utilize dual sprockets for double 50 chain. 6x6 idler shafts utilize single sprockets for single 60 chain.

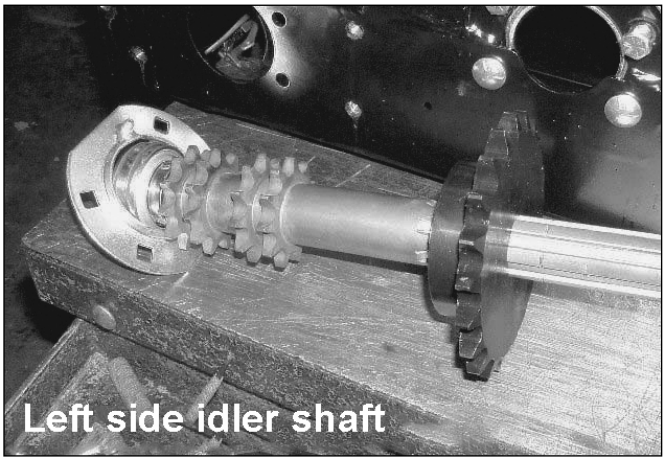
1. Install a Retaining Ring to each Idler Shaft, temporarily placing it on as far as it will go (past all retaining grooves).

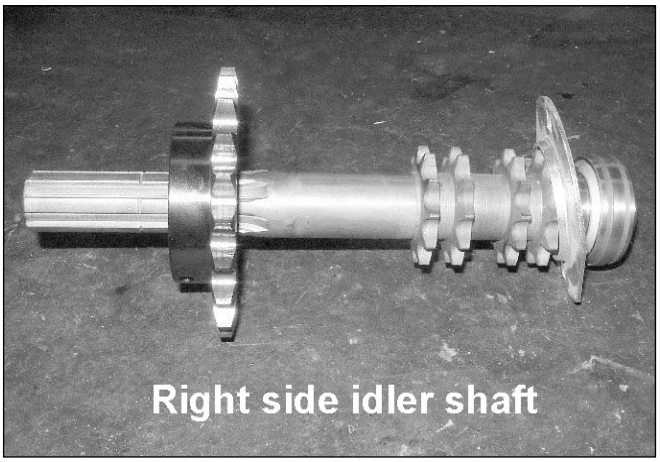
2. Install one Idler Sprocket to the shaft with the sprocket's collar facing down (Left hand idler shaft).
3. Install one Idler Sprocket to the shaft with the sprocket's collar facing up (Right hand side idler shaft).
4. Place a second Retaining Ring to the other side of the sprockets.





5. Next, install a Greasable Flange followed by a Bearing to the opposite end of the shafts.



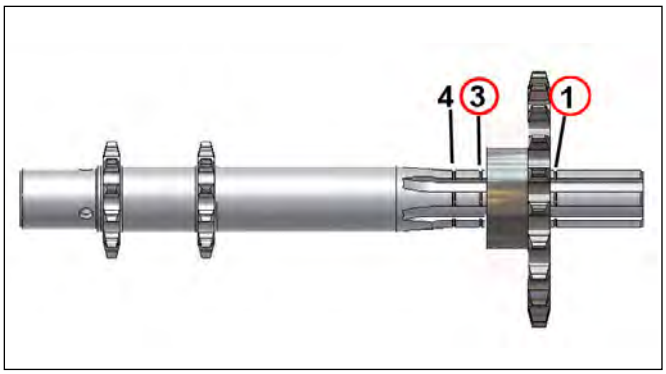


6. Install the Inner Idler Shaft Bearing Flange, Bearing and Greasable Flange assembly to the inner frame location. Loosely attach the assembly with (4) Locknut, Nylon 3/8-16, Flanged and Flat Washers.
7. Orient the 90 degree greasable flange fitting to the top and facing forward on the left hand side idler shaft and to the top facing forward on the right hand side idler shaft.

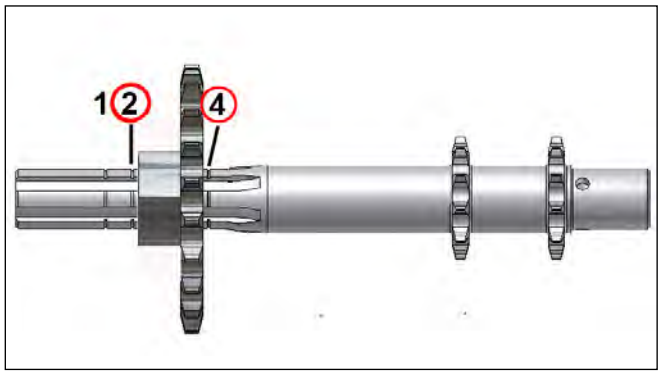




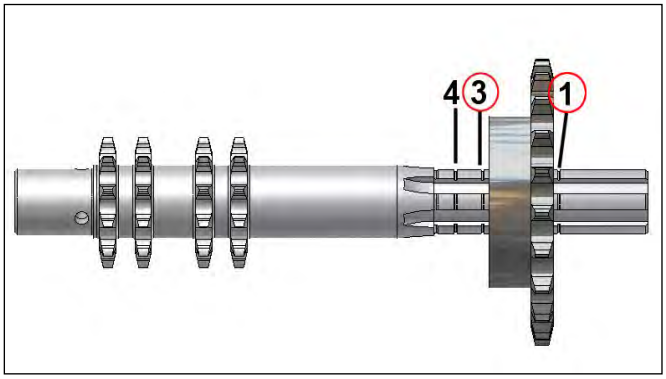
8. Rotating the bearing up in the inner flange assembly, insert the shaft into the bearing as far as it will go and then rotate down to slide back into the outer bearing.
9. Snap retaining rings into position as illustrated in the diagrams.



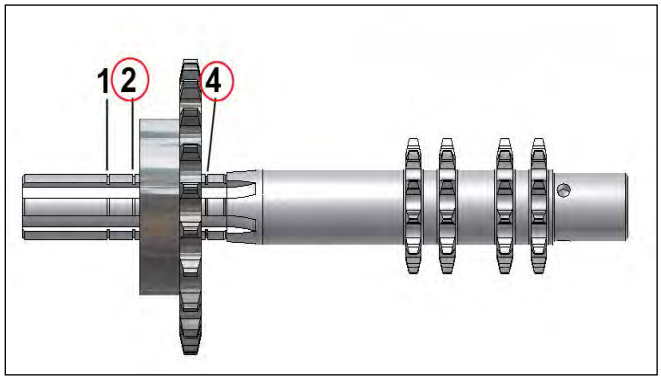
Frontier 6x6 Idler Shaft - Left



Frontier 6x6 Idler Shaft - Right



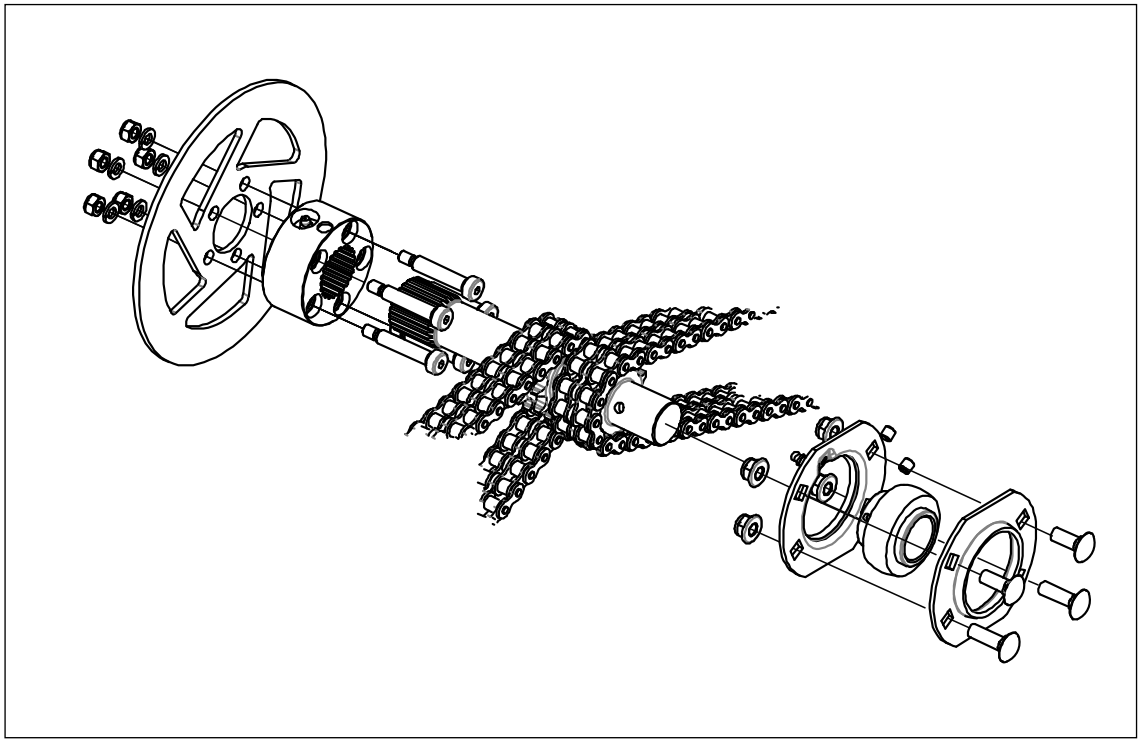
Frontier 8x8 Idler Shaft - Left



Frontier 8x8 Idler Shaft - Left

10. Align set screws with Idler Shaft countersink and install. Apply LOCTITE blue to set screws and secure tightly.
11. Tighten down all flange hardware and **torque to 35FT. LBS (47.4Nm)**
12. Perform the servicing procedure, **Drive Chain Installation.**
13. Perform the servicing procedure, **Installing the Steering Brake Disc (Frontier) .**
14. Perform the servicing procedure, **Installing The Idler Chain - Manual Adjust (Frontier) or Installing The Idler Chain-Self Adjust (Frontier).**
15. Perform the servicing procedure, **Installing The Hydraulic Steering Brake Caliper - Frontier Models.**
16. Perform the servicing procedure, **Installing the Driven Clutch**, (Left side idler shaft only).
17. Perform the servicing procedure, **Installing the Front Floor Pan.**
18. Perform the servicing procedure, **Installing the Firewall**

Removing The Idler Shaft (Aurora Models)



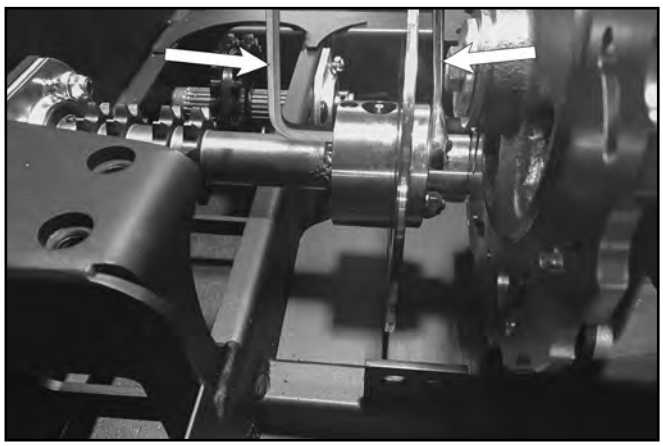
If you are removing the LEFT side Output Shaft, you will need to remove both Drive Belt and Driven Clutch to successfully work on and remove the Output Shaft.

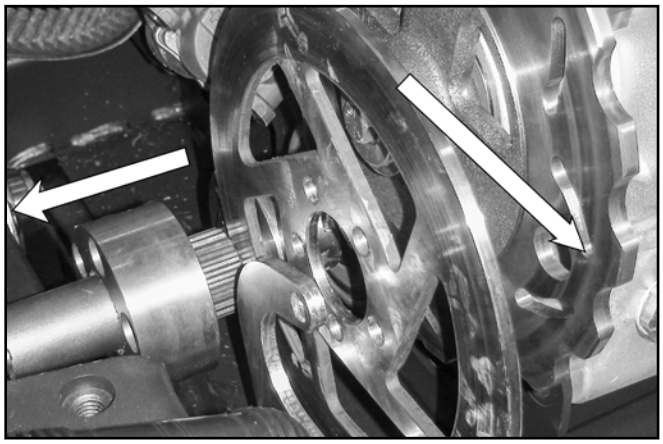
NOTE

The Brake line does NOT need to be disconnected from the caliper.

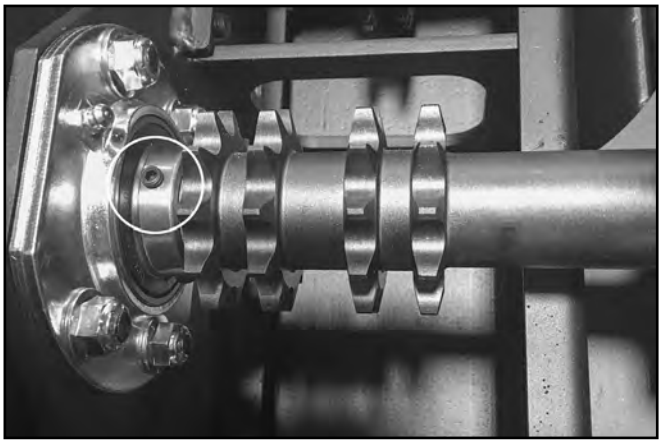
1. Perform the servicing procedure, **Removing The Firewall.**

2. Perform the servicing procedure, **Removing The Drive Belt** (if the left side Output Shaft is being serviced).
3. Perform the servicing procedure, **Removing The Driven Clutch** (if the left side Output Shaft is being serviced).
4. Perform the servicing procedure, **Removing The Hydraulic Brake Caliper - Service Brakes (Aurora)** *Do NOT disconnect the brake lines from the caliper.*
5. Perform the servicing procedure, **Removing A Drive Chain** on the front and mid chain.
6. Remove all six (5) Bolt, Shoulder M10x45 securing the Service Brake at the coupler. They are secured by six (5) Lock washer 5/16 Split-Ring Bnz and six (6) Locknut, Nylon M8x1,25mm Zn8.
7. Slide the coupler back on the output shaft and slip out the Service Brake Disc.





8. Locate the outer Idler Shaft Bearing and remove the set screw.
9. Loosen the flange hardware.
10. Push the idler shaft in towards the inner bearing and tilt up to slip the shaft out from the bearing.

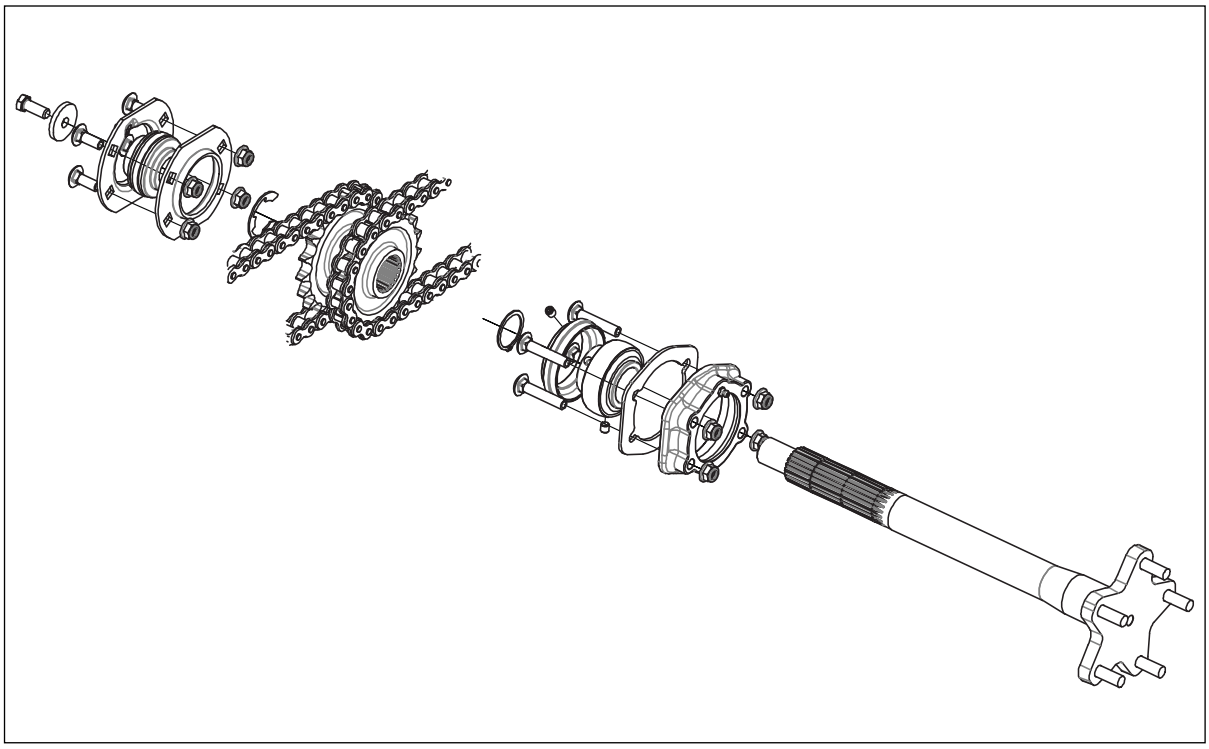


Installing The Idler Shaft (Aurora Models)

1. With the outer bearing flange hardware loosely installed, tilt the outer bearing up while inserting the idler shaft into the bearing as far as it will go.
2, Tilt the Idler Shaft assembly back down until the couplers mate at the transmission.
3. Reinstall the coupler bolts and lock washers and **Torque to 22 FT.LBS. (30.0 Nm).**
4. Perform the servicing procedure, **Installing A Drive Chain** on the front and mid chain.
5. Perform the servicing procedure, **Installing The Hydraulic Brake Caliper - Service Brakes (Aurora)**

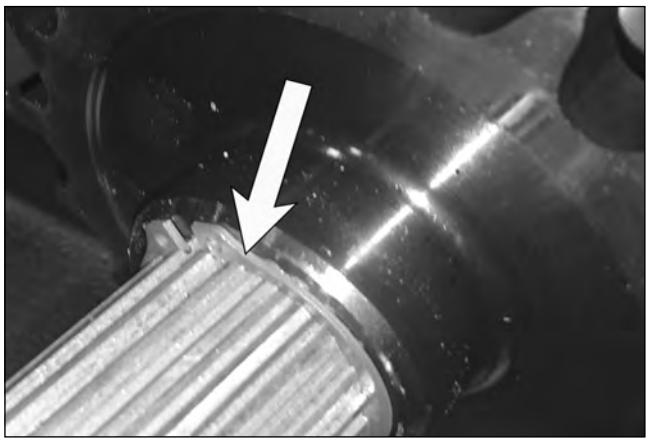
6. Perform the servicing procedure, **Installing The Driven Clutch** (if the left side Idler Shaft was being serviced).
7. Perform the servicing procedure, **Installing The Drive Belt** (if the left side Idler Shaft was being serviced).
8. Perform the servicing procedure, **Installing The Firewall.**

Removing An Axle/Sprocket/Bearing (Without Bearing Extension)



1. Locate the axle/sprocket/bearing to be removed and perform the servicing procedure, **Removing the Drive Chain.**
2. Locate the bolt threaded to end of the axle at the inner bearing and remove.
3. Remove the Retaining Ring, from the groove on the shaft to allow the sprocket to float freely on the shaft.



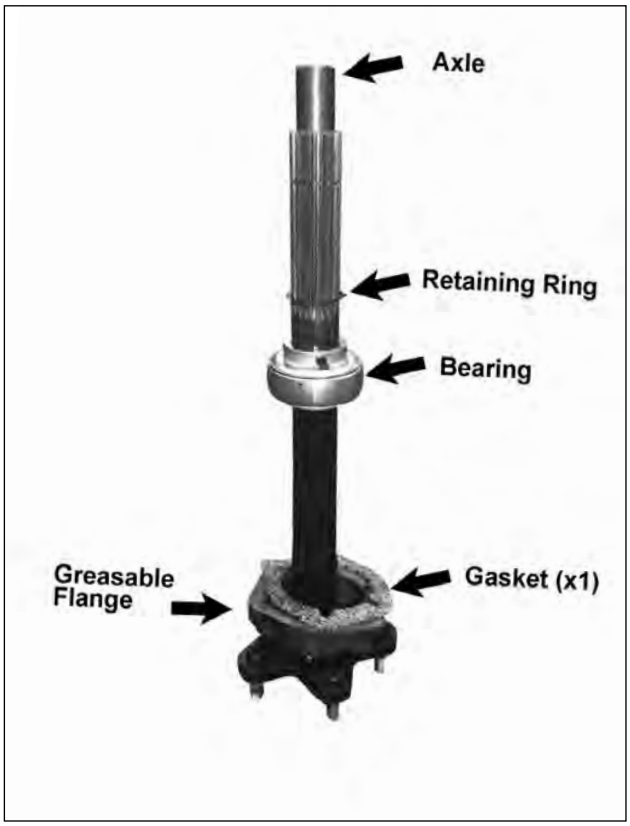


4. Remove the tire from the axle.

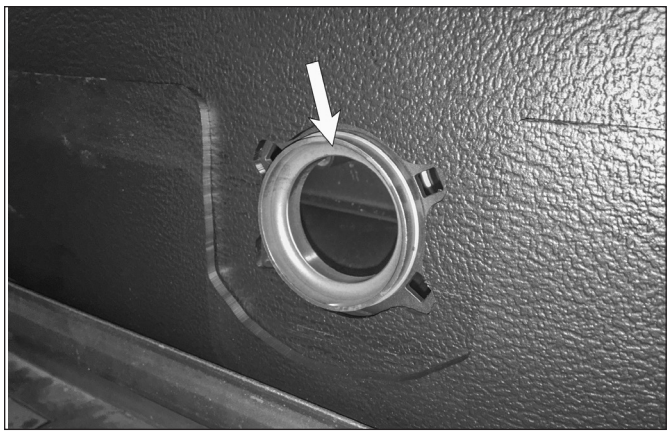
5. Remove all hardware securing the flanges of both inner and outer bearings.
6. Pull the axle from the vehicle. If necessary use a soft faced mallet to knock the axle at the hub, from behind.
7. Retrieve the drive sprocket(s) from inside the vehicle once the axle has been completely removed from the vehicle.
8. Remove the axle assembly to a clean work bench for further disassembly
9. Remove gaskets and flanges.
10. Remove the set screw(s) of the outer bearing and slide the bearing from the axle. If the bearing does not slide off freely, an arbor press may be needed.
11. Remove the greasable flange.

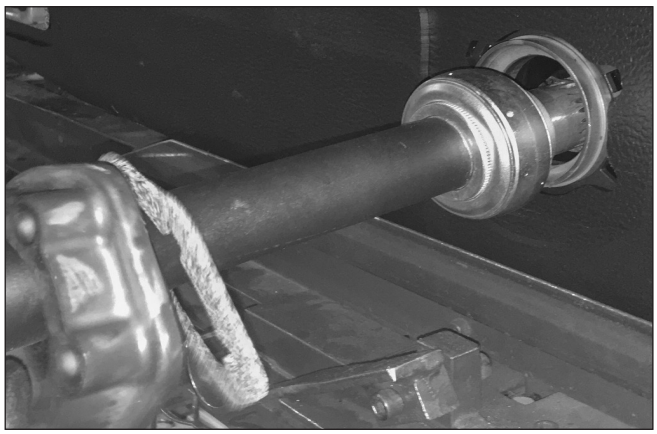
Installing an Axle/Bearing/Sprocket (Without Bearing Extension)

1. Loosely install the inner bearing/flange assembly to the vehicle frame.
2. Locate the axle and stand it upright on a clean work bench.
3. Slip the outer bearing greasable flange over the axle shaft along with one (1) cork gasket.
4. Slip the outer bearing over the axle shaft, orienting it with the collar facing up.
5. Align the flat machined surface closest to the *sprocket* end of the axle, with the set screw hole of the bearing. Apply 242 blue LOCTITE to the threads of the set screw and install. Snug the set screw to the shaft. Do NOT over-tighten.



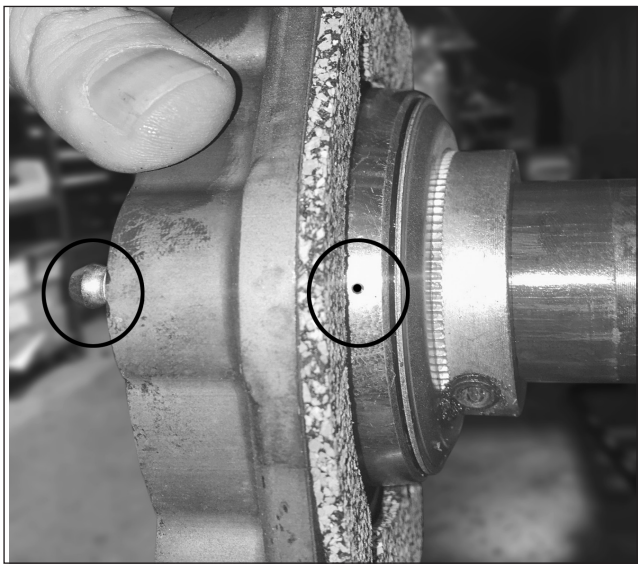
6. Temporarily install a retaining ring over the shaft as close to the bottom of the spline as possible.
7. Place the bearing carrier into axle hole location on the frame and install the assembled axle through the lower body.



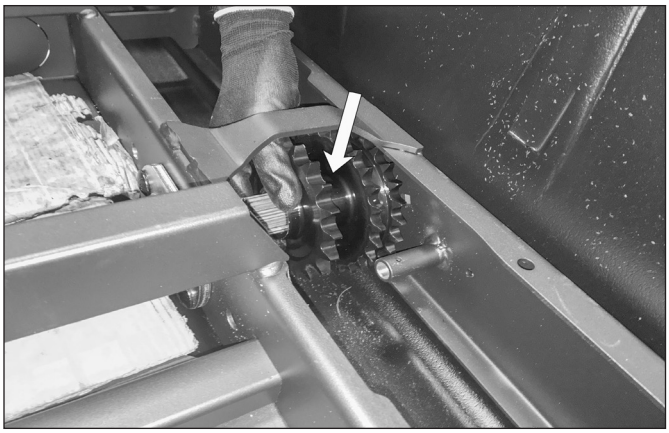


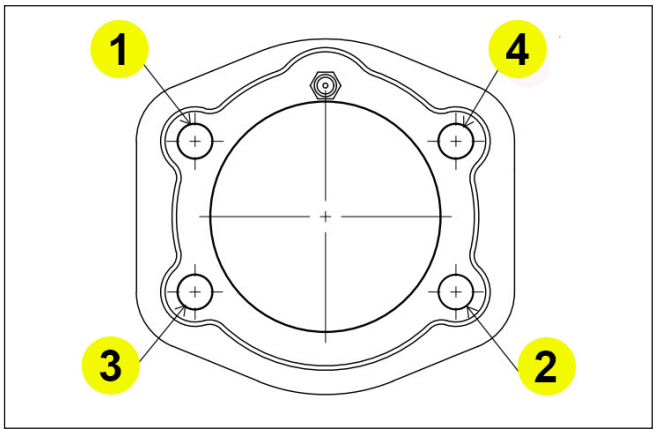
IMPORTANT

Align the grease zerk on the bearing flange with the grease hole in the bearing while installing the axle to the vehicle



8. On the inside of the vehicle, slip on each sprocket assembly orienting the sprockets as illustrated in the **Drive Chain and Sprocket Orientation Charts.**
9. Insert the axle into the inner bearing.
10. Secure the outer greasable flange with four (4) Locknut, Nylon 3/8-16, Flanged and **pre-torque to 17 FT.LBS. (23.05Nm)** in a cross corner pattern. **Final Torque to 35 FT.LBS (40.6 Nm)**





11. Tighten down inner bearing flange hardware and **torque to 30 FT.LBS (40.6 Nm)**

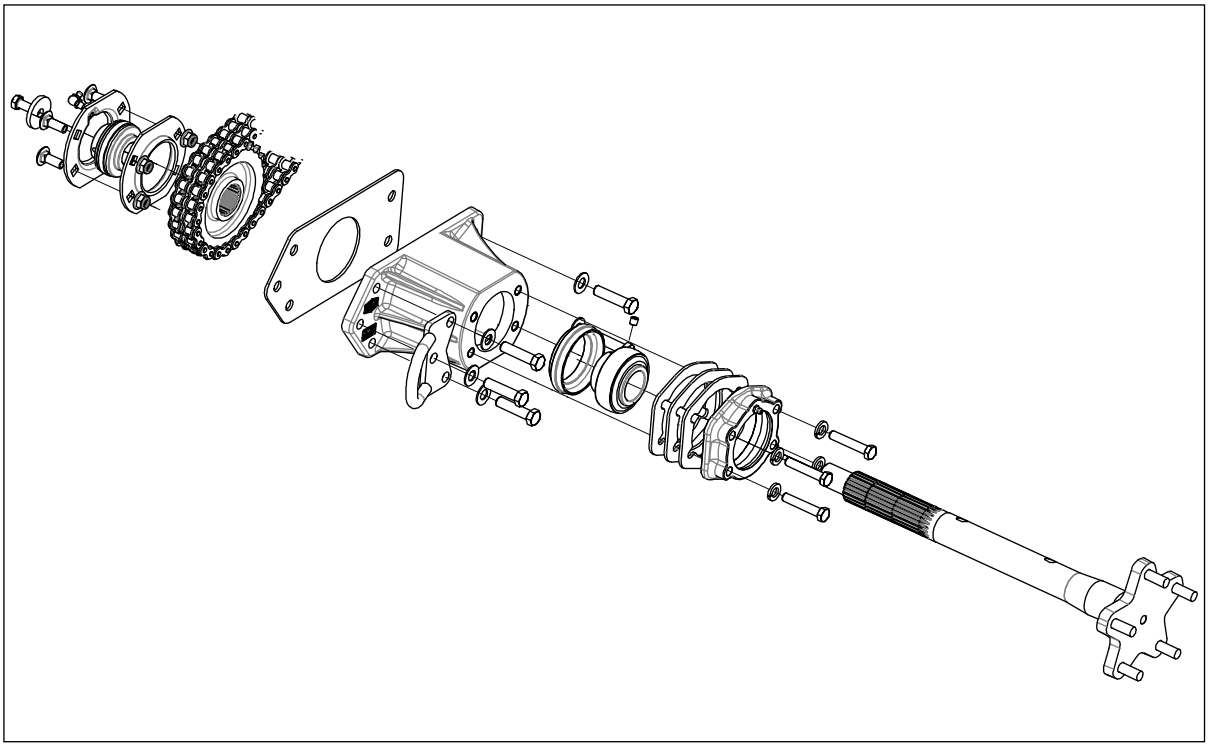
12. Align each of the final drive sprockets as shown in the **Drive Chain and Sprocket Orientation Charts**. Ensure retaining clips are snapped into the grooves on the axles and positioned on each side of the sprocket(s) assemblies to minimize flotation along the shaft under vehicle application.

IMPORTANT

Front wheel drive sprockets on all vehicle models are free floaters and do not utilize retaining rings on either side of sprocket.

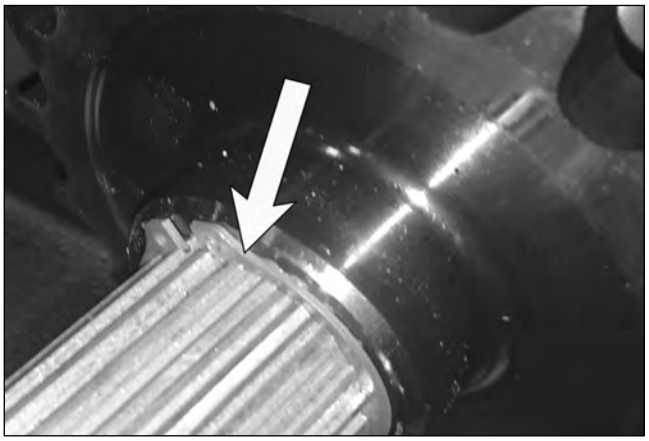
13. Install the bolt and flat washers to the end of the axle. Apply blue 242 LOCTITE to the threads of the bolt and **torque to 35 FT. LBS.(47.5Nm).**
14. Perform the servicing procedure, **Outer Axle & Bearing Lubrication.**
15. Perform the servicing procedure, **Inner Axle Bearing Lubrication.**
16. Perform the servicing procedure, **Installing A Drive Chain.**

Removing an Axle/Bearing/Sprocket (With Bearing Extension)



1. Locate the Front or Rear axle/sprocket/bearing to be removed and perform the servicing procedure, **Removing A Drive Chain(s)**
2. Locate the bolt threaded to end of the axle at the inner bearing and remove. *See following page.*
3. Remove the Retaining Ring, from the groove on the shaft to allow the sprocket to float freely on the shaft. *See following page.*

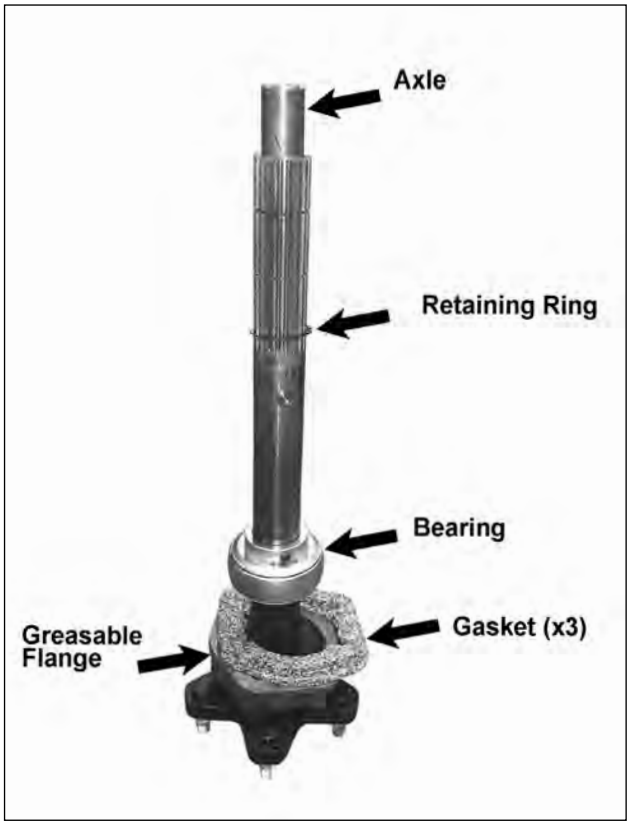




4. Remove the tire from the axle.
5. Remove all hardware securing the flanges of both inner and outer bearings.
6. Pull the axle from the vehicle. If necessary use a soft faced mallet to knock the axle at the hub, from behind.
7. Retrieve the drive sprocket(s) from inside the vehicle once the axle has been completely removed from the vehicle.

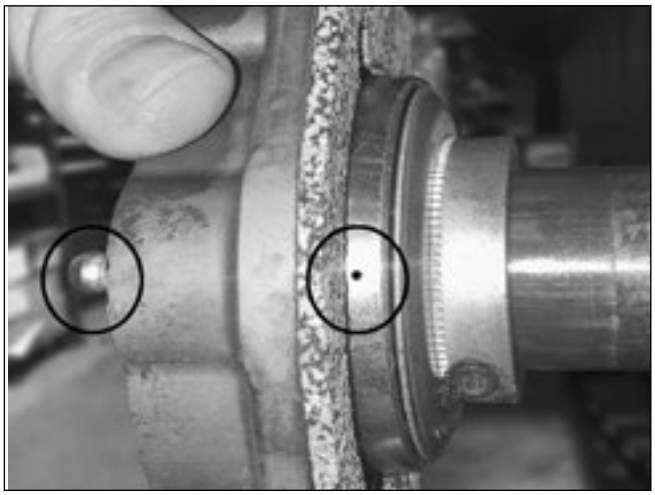
Installing an Axle/Bearing/Sprocket (With Bearing Extension)

1. Loosely install the inner bearing/flange assembly to the vehicle frame.
2. Locate the axle and stand it upright on a clean work bench.
3. Slip the outer bearing greasable flange over the axle shaft along with three (3) cork gaskets.
4. Slip the outer bearing over the axle shaft, orienting it with the collar facing up.
5. Align the flat machined surface closest to the *wheel hub* end of the axle, with the set screw hole of the bearing. Apply 242 blue LOCTITE to the threads of the set screw and install. Snug the set screw to the shaft. Do NOT over-tighten.

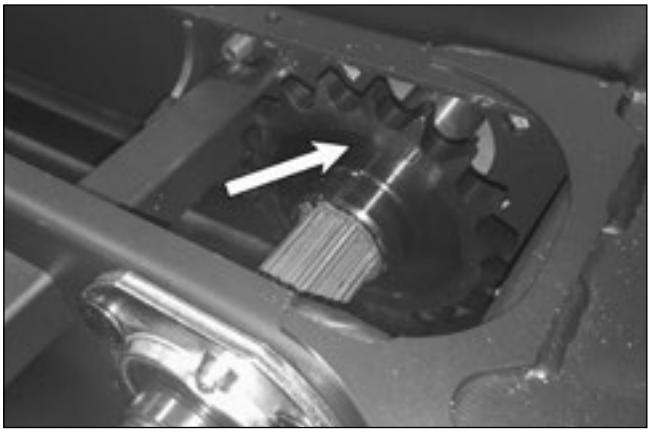


6. Temporarily install a retaining ring over the shaft as close to the bottom of the spline as possible.
7. Install the assembled axle through the lower body at the bearing extension.

8. Align the grease zerk on the bearing flange with the grease hole in the bearing while installing the axle to the vehicle
9. On the inside of the vehicle, slip on the sprocket assembly orienting the sprockets as illustrated in the **Drive Chain and Sprocket Orientation Charts.**







10. Secure the outer greasable flange with four (4) HHCS 3/8-16x2.0 Gr5 Zn and Flat Washers and **torque to 30 FT.LBS (40.6 Nm).**
11. Tighten down inner bearing flange hardware and **torque to 30 FT.LBS (40.6 Nm)**
12. Align each of the final drive sprockets as shown in the **Drive Chain and Sprocket Orientation Charts**. Ensure retaining clips are snapped into the grooves on the axles and positioned on each side of the sprocket assemblies to minimize flotation along the shaft under vehicle application.

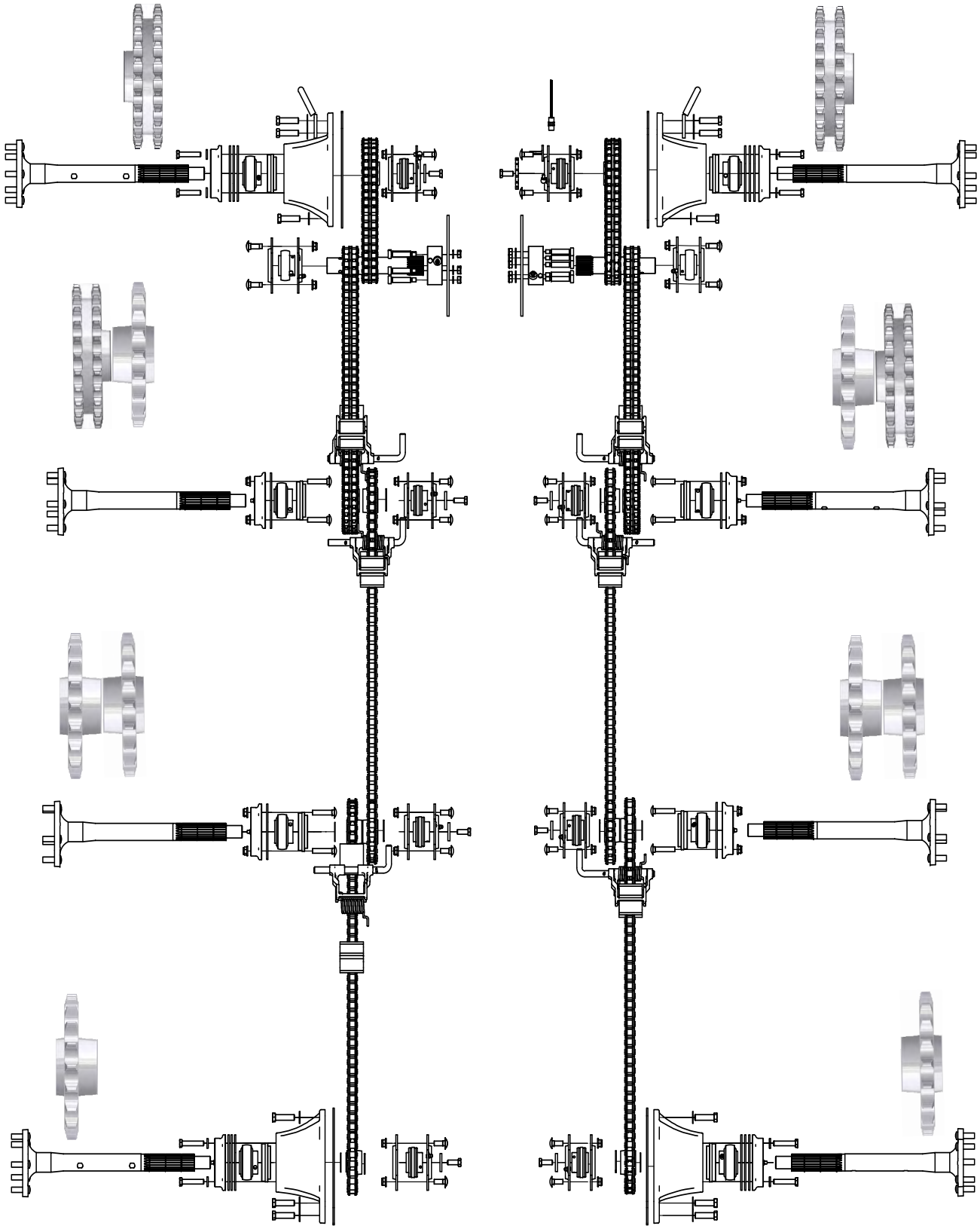
IMPORTANT

Front wheel drive sprockets on all vehicle models are free floaters and do not utilize retaining rings on either side of sprocket.

13. Install the bolt and flat washers to the end of the axle. Apply blue 242 LOCTITE to the threads of the bolt and **torque to 35 FT. LBS.(47.5Nm).**
14. Perform the servicing procedure, **Outer Axle & Bearing Lubrication.**
15. Perform the servicing procedure, **Inner Axle Bearing Lubrication.**
16. Perform the servicing procedure, **Installing A Drive Chain.**

Drive Chain and Sprocket Orientation (8x8 Models)*

*Bearing Extensions may or may not be present depending on model.



Drive Chain and Sprocket Orientation (6x6 Models) *

*Bearing Extensions may or may not be present depending on model.

